Structured questions

|!|BCELA0010000001|!| ★★ **SI** **NOS**

In the past, scurvy was a common disease among sailors and its cause was unknown. A doctor called James Lind designed an experiment to investigate the cause of scurvy.

During one of his sailing trips, he divided 12 sailors suffering from scurvy into six pairs and gave additional food on their basic diet. He then observed whether the pairs recovered from scurvy. He maintained the same environmental conditions and diet among all pairs. The type of additional food and the results are tabulated as below:

Pair	Additional food	Results
1	cider	Symptoms did not disappear.
2	sulphuric acid	Symptoms did not disappear.
3	seawater	Symptoms did not disappear.
4	garlic, mustard	Symptoms did not disappear.
5	vinegar	Symptoms did not disappear.
6	oranges and lemons	Symptoms disappeared.

(a) How did James Lind manipulate the control variables? (2 marks)

(b) Lind drew a conclusion that a lack of citric fruits, such as oranges and lemons in the diet could cause scurvy. Suggest why the design of his experiment was not adequate to demonstrate a lack of citric fruits in diet could cause scurvy. (3 marks)

(c) Scientific knowledge relies heavily on rational arguments and skepticism. After reading the story of James Lind, a scientist was skeptical about his conclusion and wrote down the following argument:

'Citric fruits should be the cure of scurvy. It is not reasonable to conclude that a lack of citric fruits is a cause of scurvy.'

Justify the scientist's claim. (3 marks)

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##

(a) He allocated same number of sailors in each pair. (1)
He maintained the same environmental conditions and diet among all pairs. (1)

(b) The sample size of each group is not large enough. (1) /
Feeding sulphuric acid or seawater to sailors would cause health problems, which may deteriorate scurvy. (1) /
There is no controlled experiment, i.e. pairs of sailors having normal diet. (1) /
Sailors were not randomly allocated into each group receiving different additional food. (1) (Max. 3)

(c) The scientist's claim is correct. (1)
Symptoms in sailors disappeared after having citric fruits, showing that citric fruits could be a cure of scurvy. (1)
There was no evidence on the deficiency of citric fruits of sailors shown. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Applying concepts	3
(c)	Critical thinking	3

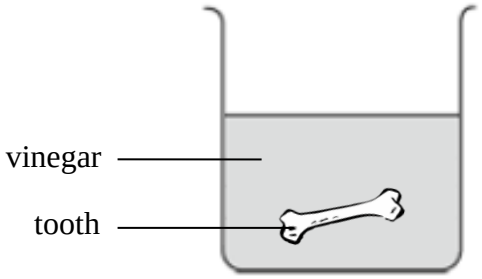
|!|BCELA0010000002|!| ★★ **SI**

Tooth decay is caused by the action of bacteria in our mouth. After a meal, the bacteria and food debris form a sticky and invisible layer called plaque on the surface of the teeth. Bacteria in the plaque act on the sugar in food debris to produce acid. The acid will dissolve the enamel and make a hole. Tooth decay begins.

(a) Describe one method to show that the bacteria in plaque are living organisms.
(4 marks)

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(b) The student carried out an experiment to investigate the action of plaque bacteria. He used vinegar to simulate the acid produced by plaque bacteria and a piece of chicken bone to simulate a tooth. The diagram below shows the set-up.



(i) State the assumption that should be made in order to make a valid investigation. (1 mark)

(ii) Suggest a control set-up for the above experiment. (1 mark)

(c) Consider the pH value of saliva, explain the role of saliva in protecting our teeth from plaque bacteria. (2 marks)

##

(a) Any ***one set*** of the following: (Max. 4)

Characteristic tested	Testing method	Expected observation
Growth (1)	• Culture plaque bacteria on an agar plate for a period of time (1) • Observe the change in the number of plaque bacteria (1)	Increase in number of bacteria / colonies (1)
Respiration (1)	• Culture plaque bacteria in a closed system, e.g. a stoppered test tube or a syringe, for a period of time (1) • Use hydrogencarbonate indicator or limewater, to test for carbon dioxide, a product of respiration (1)	Hydrogencarbonate indicator turns from red to yellow / limewater turns milky (1)

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Nutrition (1)	• Add a known amount of glucose solution to the plaque bacteria culture (1) • Detect the change in the amount of glucose with the Benedict's test (1)	The amount of glucose decreases (less brick-red precipitate is formed in the Benedict's test) (1)

(b) (i) The acid produced by the plaque bacteria is the only cause of enamel damage. (1)

(ii) Leave a chicken bone exposed to air. / Immerse a chicken bone in distilled water. (1)

(c) Saliva is slightly alkaline, (1)

which helps neutralize the acid produced by plaque bacteria. (1)

(8 marks)

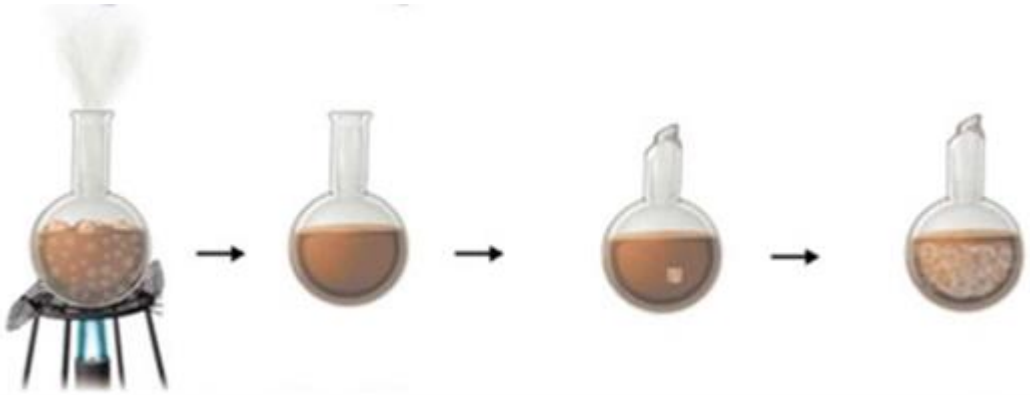
##

	Skills assessed	Marks
(a)	Applying concepts	4
(b)(i)	Practical skills	1
(b)(ii)	Practical skills	1
(c)	Applying concepts	2

|!|BCELB0010000003|!| ★ **SI**

For centuries, many people believed in spontaneous generation, the idea that living organisms could arise from non-living substances.

In 1745, John Needham carried out the following experiment.



The broth was briefly boiled.

The flask was left open for the broth to cool.

The opening of the flask was sealed.

Microorganisms grew in the broth after a few days.

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(a) Needham concluded that microorganisms had arisen from the non-living substances in the broth. How did Needham arrive at this conclusion? (2 marks)

_____

_____

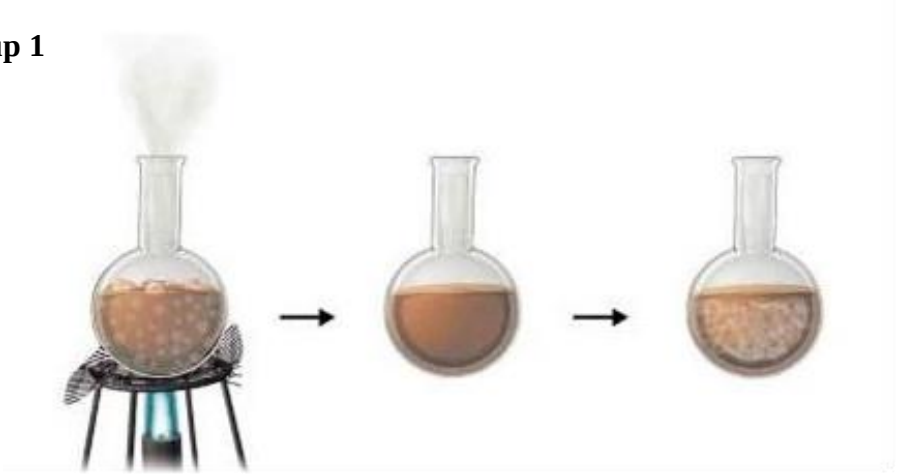
(b) However, Needham's findings were suspected to be due to experimental errors. Give ***one*** possible error in Needham's experiment. (1 mark)

_____

_____

In 1767, Lazzaro Spallanzani modified Needham's experiment as follows.

Set-up 1

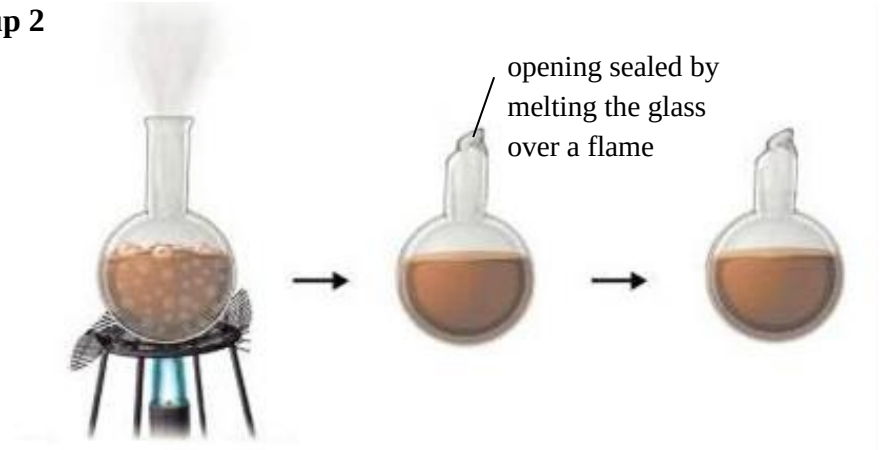


The broth was heated to boil.

The flask was left open.

Microorganisms grew in the broth after a few days.

Set-up 2



The broth was heated to boil.

The flask was sealed quickly.

No microorganisms grew in the broth even after a long time.

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(c) Suggest ***two*** control variables in set-ups 1 and 2. (2 marks)

(d) While Spallanzani claimed his own experiment as a disproval of spontaneous generation, proponents of spontaneous generation did not agree. Suggest a reason against Spallanzani's claim. (2 marks)

##

(a) Needham reasoned that boiling had killed any living things in the broth (1)
and that the sealed opening prevented any living thing from entering. (1)
Thus, the microorganisms must have arisen from the non-living substances in the broth.

(b) Any ***one*** of the following: (1)

- The broth might not be sufficiently boiled to kill all organisms inside.
- The broth might be contaminated when the flask remained open for cooling.

(c) The same type of broth / the same length of time of boiling / the same type of flask (or other reasonable answers) (Any ***two***, 2)

(d) Spallanzani had prevented air (oxygen) from entering the flask. (1)
It was thought that air (oxygen) was necessary for spontaneous generation /
The experiment only showed that microorganisms appeared when air could come into contact with the broth. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Practical skills	1
(c)	Practical skills	2
(d)	Applying concepts	2

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|!|BCELB0010000004|!| ★ SI

A disease causes the formation of black spots on the leaves of rose bushes.



John investigated the emergence of black spots in rose bushes in two areas. He picked 30 leaves at random from rose bushes growing in area A where air pollution is severe. He counted the number of leaves with black spots as follows.

Number of spots per leaf	Number of leaves
0	3
1	3
2	5
3	8
4	7
5	3
6	1

Table 1

John also picked 30 leaves at random from rose bushes growing in area B where the air is clean. The table below shows the results.

Number of spots per leaf	Number of leaves
3	1
4	1
5	2
6	3
7	8
8	9
9	6

Table 2

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(a) Construct a bar chart to show the data in **Table 1** and **Table 2**. (5 marks)

(b) With reference to the results, describe the effect of air pollution on this disease. (2 marks)

(c) Suggest ***one*** way to increase the reliability of the above investigation. (1 mark)

##

(a) Correct title (1)
Correct choice of axes (1)
Axes with labels and units (1)
Correct drawing of bars for the two sets of data (1, 1)

(b) More spots in non-polluted area. Reference to data, e.g. most leaves had 3 or 4 spots in area A, while most leaves had 7 to 9 spots in area B. (1)
The disease is less widespread in the area with more air pollution.
(or vice versa) (1)

(c) Any ***one*** of the following: (1)
- larger sample / collect more leaves
- compare leaves of same age / same size / same position on the plants
- all leaves from the same species

(or other reasonable answers)

(8 marks)

##

	Skills assessed	Marks
(a)	Presenting information	5
(b)	Analysing data	2
(c)	Practical skills	1

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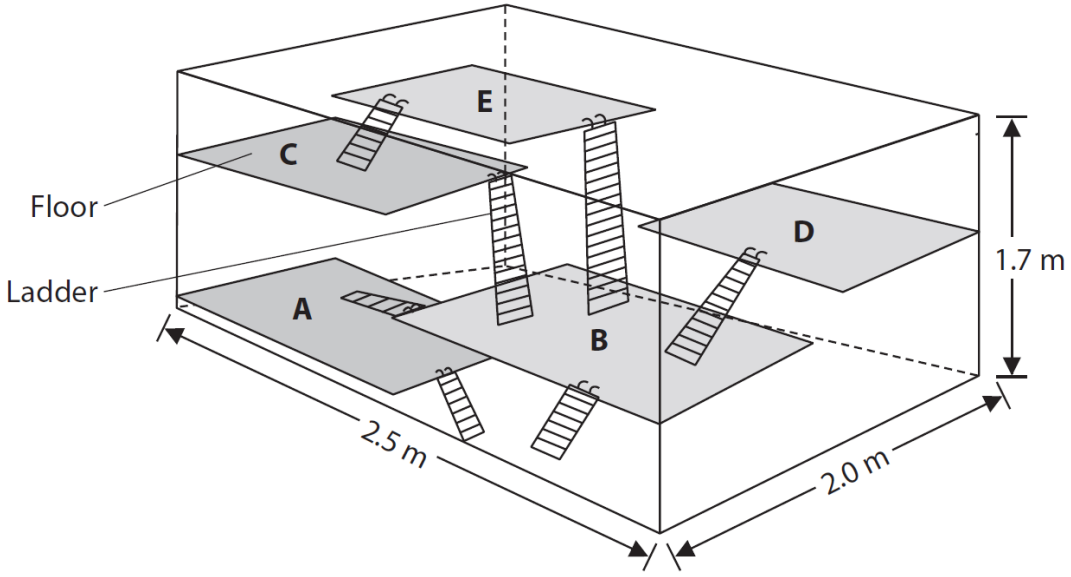
|!|BCELA0010000005|!| ★★ **Cross topic** **SI**

Edexcel GCE AL Biology Unit 5 Jun 2012 Q3

(a) An investigation was carried out to study the ability of rats to learn. A number of rats were divided into two groups, P and Q.

The rats in group P were deprived of food for twenty hours and then released into a cage. The cage contained hidden food and the rats were left in this cage for four hours each day. This was repeated each day for fourteen days.

The diagram below shows the cage. In the cage, the floors A, B, C, D and E had hidden food, water, wooden blocks, freshly cut wood chips, branches, fresh leaves, plastic containers and paper bags.

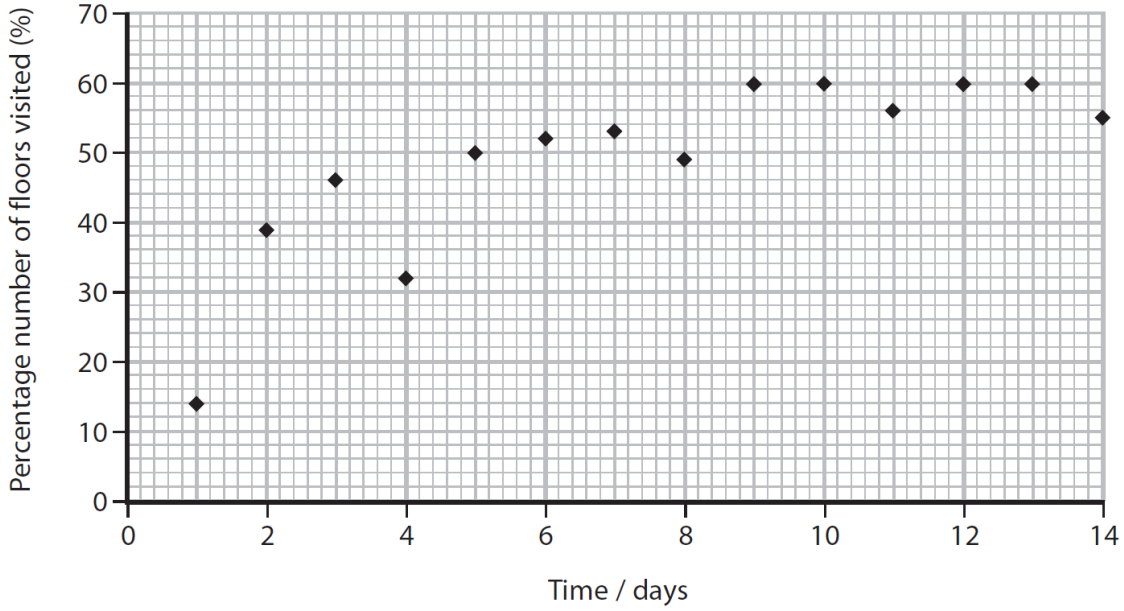


(i) The rats in group Q were used as a control. Describe how the rats in control group Q would have been treated. (2 marks)

(ii) Explain why the rats were not fed for twenty hours each day. (1 mark)

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(b) During each four-hour period in the cage, the number of floors visited by the rats in group P was recorded as a percentage of the total number of floors. The graph below shows the results of this experiment.



Using the information in the graph, describe the behaviour of the rats in group P over the fourteen-day period during this investigation. (3 marks)

(c) In a second experiment, the two groups of rats were placed in a maze containing hidden food. The percentage of rats from each group that found the food in a short period of time was recorded. The results are shown in the table below.

Group	Percentage of rats finding food (%)
P	85
Q	0

Explain the effect of the first experiment on the ability of rats to find food in a short period of time. (2 marks)

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(d) The brains of both groups of rats were examined. The mean spine density per neurone for each group was calculated.
Spine density represents a measure of the number of synapses per neurone.
The results are shown in the table below.

Group	Mean spine density / arbitrary units
P	1.96
Q	1.78

Suggest how these results explain the effect of the first experiment on the ability of rats to find food using the cage. (2 marks)

_____

_____

##

(a) (i) Group Q should be put in the same cage but with no enrichment. (1)
They should be treated the same as group P, e.g. same length of starvation time, feeding time, time kept in the cage, etc. (1)

(ii) Starvation encourages them to look for food. (1)

(b) The percentage number of floors visited increases. (1) /
Rapid increase occurred during the first five days. (1) /
After the 5th day, the increase slowed down / levelled off. (1) /
The rats did not visit all the floors (on each day), e.g. 100% of the floors never achieved. (1) (Max. 3)

(c) In the first experiment, hidden food in the cage encouraged the rats to explore the environment. (1)
They learned to look for food, so they visit more of the maze. (1)

(d) More synapses allow more connections between neurones. (1)
The learning capacity of group P improved. (1)

(10 marks)

##

	Skills assessed	Marks
(a)(i)	Practical skills	2
(a)(ii)	Applying concepts	1
(b)	Analysing data	3
(c)	Analysing data	1
	Applying concepts	1
(d)	Analysing data	1
	Applying concepts	1

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|!|BCELC0010000006|!|

Excretion is a common characteristic of living organisms.

(a) State ***two*** other common characteristics of living organisms. (2 marks)

_____

_____

(b) Like other living organisms, plants also need to excrete metabolic wastes. Describe how the waste product produced in respiration is removed from a leaf. (2 marks)

_____

_____

(c) During the cold winter, some plants shed their leaves. This can be described as an excretory mechanism. Suggest ***two*** other reasons why plants shed their leaves during winter. (2 marks)

_____

_____

(d) State ***one*** excretory organ in humans, and the metabolic waste it excretes. (2 marks)

_____

_____

##

(a) Any ***two*** of the following: (2)
nutrition / respiration / movement / sensitivity / growth / reproduction

(b) The carbon dioxide produced in respiration diffuses out (1)
of the leaf through the stomata (along the concentration gradient). (1)

(c) To reduce water loss (1)
To conserve energy (1)
(Or other reasonable answers)

(d) Any ***one set*** of the following: (2)

- The lungs (1) excrete carbon dioxide. (1)
- The kidneys (1) excrete urea. (1)
- The skin (1) excretes urea. (1)

(8 marks)

##

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	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	2

|!|BCELB0010000007|!| ★ **SI**

Studying biology involves carrying out scientific investigations.

(a) Scientific investigation involves several steps as shown below.

Making a hypothesis	Analysing the results
Making observation	Carrying out an experiment

(i) Arrange the above steps of scientific investigation in the correct order. (2 marks)

(ii) What will biologists do if the experimental results

(1) support the hypothesis? (1 mark)

(2) do not support the hypothesis? (1 mark)

(b) Amy works in a small and crowded office with many colleagues. She finds that her potted plant in her office grows larger than the one in her home. She wants to find out why this occurs by carrying out a scientific investigation.

(i) What is Amy's observation in her scientific investigation? (1 mark)

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(ii) Suggest how Amy may carry out the following steps. (3 marks)

1. Making a hypothesis:

2. Making a prediction:

3. Conducting an experiment:

(iii) State ***two*** control variables in her scientific investigation. (2 marks)

##

(a) (i) 1. Making observation
2. Making a hypothesis
3. Carrying out an experiment
4. Analysing the results
(Maximum 2 marks, deduct 1 mark for each wrong answer) (2)

(ii) (1) If the experimental results support the hypothesis, the biologists will repeat the experiment many times for confirmation. (1)

(2) If the experimental results do not support the hypothesis, the biologists have to revise the hypothesis. (1)

(b) (i) She observes that her potted plant in the office grows larger than that in her home. (1)

(ii) 1. Making hypothesis: Plants grow better in an environment with more people (or higher carbon dioxide concentration). (1)

2. Making a prediction: If a plant is grown in an environment with more people, then it will grow larger than a plant grown in an environment with fewer people. (1)

3. Conducting an experiment: Grow five plants in five rooms with different numbers of people (e.g. 0, 2, 4, 6 and 8 people) (1)

(iii) Any ***two*** of the following: (2)
species of the plant / light intensity / nutrient content in the soil
(Or other reasonable answers)

(10 marks)

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##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	2
(a)(ii)	Understanding basic concepts	2
(b)(i)	Applying concepts	1
(b)(ii)	Applying concepts	3
(b)(iii)	Practical skills	2

|!|BCELA0010000008|!| ★★ **SI** **NOS** **STSE**

In the past, doctors believed that gastric ulcer was caused by excessive secretion of stomach acid induced by stress. However, many patients developed gastric ulcer again after being treated with medicines for neutralizing stomach acid. In the 1980s, the Australian doctor Barry Marshall observed that patients with gastric ulcers had a type of bacterium called *Helicobacter pylori* in their stomach. He therefore put forward a hypothesis about gastric ulcer. Based on this hypothesis, he treated his patients with antibiotics. Many of the patients recovered and did not develop gastric ulcer again.

(a) Suggest the hypothesis made by Marshall about gastric ulcer. (1 mark)

_____

_____

(b) Based on the hypothesis you answered in (a), describe how you would test whether it is correct or not. (2 marks)

_____

_____

_____

(c) Explain why many doctors were surprised about Marshall's observation. (1 mark)

_____

_____

Scientists study how *Helicobacter pylori* survives in the stomach. To avoid the acidic environment of the stomach, *H. pylori* uses its flagella to burrow into the mucus lining of the stomach to reach the epithelial cells underneath, where it is less acidic. *H. pylori* is able to detect the pH gradient in the mucus and move towards the less acidic region.

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(d) Give ***two*** characteristics of organisms demonstrated by *Helicobacter pylori* in the passage above. (2 marks)

(e) Elaborate on how the events above demonstrate the following aspect of the nature of science: Scientific knowledge is tentative and subject to change. (2 marks)

##

(a) *Helicobacter pylori* causes gastric ulcer. (1)

(b) Introduce *Helicobacter pylori* into the stomach of healthy mammals. (1)
If the hypothesis is correct, the mammals will develop gastric ulcer. (1)

(c) They thought that the acidic environment in the stomach is unfavourable to the survival of bacteria. (1)

(d) Movement, sensitivity (1, 1)

(e) Barry Marshall's new findings provide evidence that gastric ulcer is caused by *Helicobacter pylori*. (1)
His findings corrected the previous misconception that gastric ulcer was caused by excessive secretion of stomach acid. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Practical skills	2
(c)	Applying concepts	1
(d)	Applying concepts	2
(e)	Applying concepts	2

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|!|BCELB0010000009|!| ★ **SI** **NOS**

Read the passage below and answer the questions that follow:

Power plants emit sulphur dioxide which is a type of air pollutant. A farmer whose fields are near to a power plant thinks that the yield of soybeans is reduced because of the emission of sulphur dioxide. To find out whether this is true, scientists worked with farmers to study the effect of sulphur dioxide concentration on the growth of soybeans.

In one of their studies, the scientists grew soybean plants in a greenhouse. They divided the soybean plants into two groups:

1. The plants were subject to sulphur dioxide gas for a few hours per day to simulate the emission of sulphur dioxide from power plants.
2. The plants were left to grow in the greenhouse normally and without sulphur dioxide treatment.

The growth of the two groups of soybean plants was monitored for a year. During this period, the local scientists worked with other scientists from foreign countries on the sulphur dioxide emission issue.

(a) Identify the hypothesis of this investigation. (1 mark)

(b) Suggest ***two*** variables that the scientists can measure to represent growth. (2 marks)

(c) Identify the control group of the study. Explain the purpose of including the control group. (2 marks)

(d) What are control variables? Suggest ***two*** control variables in this study. (3 marks)

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(e) Suggest what aspects of the nature of science are reflected from the following: 'The local scientists worked with other scientists from foreign countries on the sulphur dioxide emission issue.' (1 mark)

_____

_____

##

(a) Sulphur dioxide lowers the yield of soybeans. (1)

(b) Any ***two*** of the following: (2)

- the number of bean pods in each plant
- the number of seeds per pod in each plant
- the dry mass of the pods in each plant

(Or other reasonable answers)

(c) The group that was left to grow in the greenhouse normally and without sulphur dioxide treatment. (1)

It is used to show that any reduction of the yield of soybean is due to sulphur dioxide. (1)

(d) Variables that have to be kept constant. (1)

Any ***two*** of the following: (2)

- age of the soybean plants
- species of the soybean plants
- number of soybean plants in each group
- water supply
- light intensity
- temperature

(Or other reasonable answers)

(e) Scientists collaborate with each other. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Applying concepts	2
(c)	Applying concepts	2
(d)	Understanding basic concepts	1
	Applying concepts	2
(e)	Applying concepts	1

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|!|BCELA0010000010|!| ★★ SI

The beet caterpillar is an insect pest that feeds on the leaves of tomato plants. When beet caterpillars feed on the leaves of a tomato plant, the leaves will release a chemical X. In an investigation, some scientists suggest that chemical X attracts wasps to the tomato plants attacked by caterpillars.

(a) What is the hypothesis of this investigation? (1 mark)

(b) The scientists design an experiment to test the hypothesis.

(i) Outline how the scientists would carry out the experiment. (4 marks)

(ii) Predict the experimental results that would support the hypothesis. (1 mark)

(c) Explain an advantage for the plant to release chemical X when its leaves are being eaten by caterpillars. (1 mark)

##

(a) Chemical X attracts wasps. (1)

(b) (i) Prepare two groups of tomato plants that are of the same age and state of health. (1)

One group is affected with beet caterpillars, the other is unaffected. (1)

Place both groups in the same environment (e.g. the temperature and water availability are kept constant). (1)

Release wasps and observe their activity. (1)

(ii) The hypothesis is supported when more wasps visit the plants eaten by caterpillars than the unaffected group. (1)

(c) Chemical X attracts wasps which kill / eat beet caterpillars. (1)

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(7 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)(i)	Applying concepts	4
(b)(ii)	Applying concepts	1
(c)	Applying concepts	1

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